CCNA-13 Integrated Enterprise Network Capstone
D3 Initial Network Build Standard / Baseline

Team: Cisco Team 3
Version: 1.4
Status: Added VLAN ACL policy and documented WLC connectivity limitation.

General Overview

This document establishes the network build standard for the Integrated Enterprise Network Capstone. The purpose of it is to provide a consistent baseline for device naming, management access, security, centralized services, and general network configuration. All devices should follow the same approved standard for the network. This way, it remains organized, secure, and easier for the team to test and troubleshoot.

This standard represents the team's initial configuration baseline. It may be modified as implementation, integration testing, and Packet Tracer capability testing identify necessary changes. Approved revisions become the standard for subsequent implementation, testing, and auditing.

Naming

Device Type	**Naming Standard**
Edge Router	KFP-R1
Core Router	KFP-R2
Multilayer Switches	KFP-MLS1, KFP-MLS2
Access Switches	KFP-SW1 through KFP-SW4
DHCP/DNS Server	KFP-DHCP-DNS
RADIUS/AAA Server	KFP-RADIUS
Syslog/NTP Server	KFP-SYSLOG-NTP
Internal Resource Server	KFP-RESOURCE
Wireless Access Point	KFP-AP1
Wireless LAN Controller	KFP-WLC1

Interface and Cabling Standard

Connection Type	Interface / Cabling Standard
Router ↔ Router	Routed Ethernet Connection
Router↔Multilayer Switch	Routed Ethernet connection; multilayer-switch interface configured as a Layer 3 routed port
Multilayer Switch↔Multilayer Switch	802.1Q Ethernet trunk
Multilayer Switch↔Access Switch	802.1Q Ethernet Trunk
Access Switch↔PC/Server	Ethernet access port assigned to the appropriate VLAN
Access Switch↔Lightweight Wireless AP	Ethernet connection using VLAN 99 for AP management
Wireless AP↔Wireless Client	802.11 wireless connection
Access Switch↔Wireless LAN Controller	WLC trunk connectivity is undergoing Packet Tracer compatibility testing. Final WLC/AP interface configuration will be documented following validation.

Device-Type Baseline

Device Type	Requirements
R1 Edge Router	Approved hostname; routed interfaces; single-area OSPF; NAT/PAT for outbound connectivity; SSH management; AAA/RADIUS with local fallback; configuration saved
R2 Core Router	Approved hostname; routed interfaces; single-area OSPF; SSH management; AAA/RADIUS with local fallback; configuration saved
KFP-MLS1/KFP-MLS2	Approved hostname; ip routing; assigned gateway SVIs according to Logical Topology; VLAN 99 management SVI; routed connection toward R2; 802.1Q trunks; single-area OSPF; assigned STP root primary/secondary roles; SSH/AAA management; configuration saved
KFP-SW1 through KFP-SW4	Approved hostname; VLAN 99 management SVI; 802.1Q uplink trunks; access ports assigned to appropriate VLANs; PortFast and BPDU Guard on applicable endpoint ports; port security on applicable endpoint ports; unused ports shutdown;

	SSH/AAA management; configuration saved
DHCP/DNS Server	Static server addressing per Addressing Plan; DHCP and DNS services enabled; DHCP scopes configured for applicable client VLANs; internal DNS records configured as required
RADIUS Server	Static server addressing per Addressing Plan; RADIUS/AAA service enabled; network devices configured as RADIUS clients where required; administrative and wireless authentication supported according to project requirements
Syslog/NTP Server	Static server addressing per Addressing Plan; NTP and Syslog services enabled; infrastructure devices use centralized time/logging where supported
Internal Resource Server	Static server addressing per Addressing Plan; internal service enabled as required for connectivity testing; used as protected permit/deny target
Wireless AP	Approved hostname/name where supported; management addressing in VLAN 99 according to the Addressing Plan; registration with the Wireless LAN Controller; wireless client connectivity through the approved WLAN; RADIUS-backed authentication through the WLC
Wireless LAN Controller	Management addressing in VLAN 99; wireless client interface in VLAN 50; lightweight AP management; WPA2-Enterprise authentication through centralized RADIUS; wireless DHCP connectivity; centralized NTP/Syslog where supported by Packet Tracer
PCs/Clients	Connected to assigned VLAN; DHCP used where specified by Addressing Plan; correct gateway/DNS received; used for permitted/denied connectivity testing

DNS Name Records

Hostname	**IPv4 Address**
dhcp-dns.kfp.internal	10.20.40.10
radius.kfp.internal	10.20.40.11
ntp-syslog.kfp.internal	10.20.40.12

resource.kfp.internal	10.20.40.13

Management Access

Network infrastructure will use the dedicated Management VLAN for in-band device management. Exact management addresses and interface assignments will follow the Addressing Plan.

- Remote administration must use SSH version 2 with 2048-bit RSA keys where supported. SSH version 1 and Telnet are prohibited.
- Unencrypted HTTP management must be disabled. HTTPS management is permitted when required and securely configured.
- RADIUS/AAA will provide centralized administrative authentication where supported
- Management access will be restricted according to the approved segmentation and ACL policy
- Devices of the same type will use consistent management-access settings

Switching and Routing Standard

The network uses 6 defined VLANs for IT/Admin, Finance, Wireless, General Staff, Servers, and Management. VLAN IDs, subnet assignments, gateway assignments, management addresses, and routed transit addressing will follow the Addressing Plan.

- Access-switch uplinks to multilayer switches will use 802.1Q trunking
- Connections between multilayer switches will use 802.1Q trunking
- Inter-VLAN routing will be performed by the multilayer switches using SVIs
- Gateway SVI ownership will follow the Logical Topology diagram
- Routed infrastructure links will use Layer 3 routed interfaces
- Internal dynamic routing will use single-area OSPF Area 0
- STP root bridge assignments will follow the Logical Topology and align with gateway ownership
- Required VLANs will be configured consistently across applicable switches
- Only required VLANs will be permitted across trunk links where practical

Access-Layer Hardening Standard

- Port security will be configured on applicable endpoint access ports
- PortFast will be enabled on applicable endpoints access ports
- BPDU Guard will be enabled on applicable PortFast/access ports
- Unused access ports will be administratively disabled
- Trunk ports will be explicitly configured as trunks where required
- Trunk ports must use explicit configuration with DTP negotiation disabled where supported, and an unused native VLAN (999) must be used consistently across applicable trunk links.

Services Standard

Centralized services will follow the server roles and static addressing defined in the Addressing Plan

- DHCP/DNS: DHCP will provide correct address range, default gateway, and DNS information for applicable client VLANs. DHCP relay will be configured where clients are separated from the centralized DHCP server by a Layer 3 boundary. DNS will provide internal name resolution

- RADIUS/AAA: The centralized RADIUS service will support authentication as required by the project. Device administration will use AAA
- NTP/Syslog: NTP will be established before Syslog is relied upon so centralized logs use consistent and trackable timestamps
- NAT/PAT: R1 will provide NAT/PAT for outbound connectivity to the simulated external network
- Wireless: The approved wireless architecture uses a Wireless LAN Controller (WLC) and lightweight wireless access point (AP) to support WPA2-Enterprise authentication through the centralized RADIUS server. Wireless clients will use VLAN 50 and receive addressing through DHCP. WLC and AP management will use VLAN 99. The planned WLC management address is 10.20.99.27, the wireless VLAN interface is 10.20.50.20, and the AP management address is 10.20.99.26

Device Security Standard

- Privileged EXEC access must use enable secret, and local administrative accounts must use username ... secret.
- A local admin account will be maintained as a fallback if RADIUS server is unavailable
- Console and VTY access will require authentication
- Service password-encryption must be enabled
- Admin sessions will use an inactivity timeout to limit indefinitely open sessions
- An authorized access warning banner will be configured on devices

ACL and Segmentation Standard

- Extended ACLs will enforce inter-segment access according to the segmentation plan
- Required traffic will be explicitly permitted where appropriate
- Restricted traffic will be denied according to the segmentation policy
- Access relationships will be tested in both permitted and denied directions
- ACLs must not unintentionally block required DHCP, DNS, or RADIUS communication
- ACL configuration will remain consistent with the approved VLAN and addressing design
- Exact ACL entries and placement may be refined during implementation and testing

VLAN/ACL	**Device**	**Rules**
Finance (20)	MLS1	Deny Management, IT/Admin, RADIUS, and Syslog/NTP access; permit required DHCP, DNS, and authorized resources
General Staff (30)	MLS2	Deny Finance and Management access; permit required network services and authorized traffic

Wireless (50)	MLS2	Deny restricted internal segments; permit required DHCP, DNS, and authorized traffic
SSH-MGMT-ONLY	Management VTY Lines	Permit approved management sources only; deny unauthorized SSH access
IT/Admin (10)	MLS1	Permit authorized SSH access to Management VLAN 99 and required DHCP, DNS, RADIUS, and infrastructure services
Management (99)	MLS1/MLS2	Allow authorized device administration and required infrastructure services; restrict unauthorized access from user VLANs

Note: Finance, General Staff, and Wireless ACLs are applied inbound on their respective VLAN interfaces. SSH-MGMT-ONLY is applied to VTY management access.